

Joshua Baldwin

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PROFESSIONAL SUMMARY

Software Engineer with an MS in Computer Science and a builder's instinct for interactive systems. Backend expertise in Java/Spring Boot microservices, API architecture, and cloud platforms, paired with hands-on indie game development in Unity/C#, including shipping a game jam title with custom gameplay mechanics, physics-driven interactions, and player-feedback polish. Comfortable moving across disciplines, from server-side architecture to character controls and game feel, and thrives in small, collaborative, cross-functional teams.

TECHNICAL SKILLS

Game Development: Unity, C#, Gameplay & Interaction Systems, Character Controls, Game Feel & Polish, Rapid Prototyping

Backend Systems: Java, Spring Boot, Microservices, Asynchronous Event Processing, REST APIs, Android Development

AI & Orchestration: LangGraph, Agentic Workflows, Multi-Agent Systems Integration

Platform & DevOps: Red Hat OpenShift, Kubernetes, Docker, Azure, Argo CD, GitHub Actions CI/CD

Engineering Standards: Fault-Tolerant Architecture, API Design Standards, Clean Architecture

GAME DEVELOPMENT PROJECTS

Lizard Wizard | GMTK Game Jam 2026 | itch.io, Unity/C#

Programmer & Game Designer | 3-Person Core Team | Rated 5/5 by players

- Co-programmed a top-down action game built and shipped in 48 hours for the Game Maker's Toolkit (GMTK) Game Jam, where players race to clean a haunted house and cast delaying spells before guests arrive.
- Built core gameplay systems from scratch, including player movement and a boost mechanic, a pick-up/throw item interaction system, and a mana-driven spellcasting loop tied to an in-world ritual mechanic.
- Partnered directly with a co-developer and an artist/sound designer to build the full experience, then used feedback from external playtesters to keep controls and game feel readable and fun under a hard jam deadline.
- Contributed directly to core game design (loop, pacing, and spell/mana balance) in addition to programming, helping shape the concept from pitch to shipped, playable build.

PROFESSIONAL EXPERIENCE

Arch MI | Greensboro, NC

Software Engineer II | October 2025 – Present

- Develop event-driven backend workflows and messaging integrations in Java and Spring Boot to support scalable asynchronous processing across core platform services.
- Develop, maintain, and support the enterprise Android mobile application, ensuring secure API communication, high stability, and seamless feature parity with core web services.
- Led the design of AI-driven backend infrastructure utilizing LangGraph for multi-agent orchestration of complex domain queries, aligning with modern ML pipeline architectures.
- Architect high-performance Java and Spring Boot backend services, ensuring high availability and resilience for high-traffic platforms.
- Modernize legacy web services into hybrid-cloud architecture using Azure and OpenShift, facilitating seamless integrations for external partners.
- Engineer standardized API distribution layers and client integration packages, enabling cross-functional engineering teams to seamlessly consume core platform services.

Software Engineer I | July 2023 – October 2025

- Designed and shipped an end-to-end single sign-on authentication flow from scratch using Keycloak, independently learning OAuth, OIDC, and modern identity protocols to deliver a secure, production-ready system that still supports thousands of daily users.
- Developed and maintained RESTful web services using Java and Spring Boot, ensuring high performance, scalability, and security in application architectures.
- Collaborated within an Agile software development team to design, develop, and implement web-based customer-facing software solutions, using Scrum methodologies to improve project delivery and team efficiency.
- Built dynamic internal tooling and web-based operational utilities to streamline team workflows and estimation processes.
- Used LoadRunner for performance testing to identify bottlenecks and optimize application scalability, ensuring robust performance under varying loads and conditions.
- Implemented CI/CD pipelines using GitHub Actions and Argo CD, automating deployment to Red Hat OpenShift containers and improving code quality through continuous integration and delivery practices.
- Conducted research on generative artificial intelligence and its potential applications within the mortgage insurance industry, exploring innovative solutions to enhance underwriting processes and improve risk assessment.

Software Engineer Intern | June 2022 – August 2022

- Led the design and implementation of a dynamic pointing tool for biweekly Agile refinement meetings, significantly improving team engagement and streamlining the estimation process.
- Designed a visualization tool for audit data events, leveraging SQL database integration to transform user data into actionable insights, empowering team members to optimize daily operations and improve performance.

Appalachian State University | Boone, NC

Graduate Teaching Assistant | August 2021 – May 2023

- Facilitated undergraduate computer science laboratories and graded assignments, guiding students through Java programming work and ensuring clear, constructive feedback.
- Conducted weekly office hours, addressing student inquiries related to software engineering principles, including design patterns and methodologies, to enhance understanding and application of key concepts.

EDUCATION

Appalachian State University | Boone, NC

Master of Science – MS, Computer Science | August 2021 – May 2023

Appalachian State University | Boone, NC

Bachelor of Science – BS, Computer Science | August 2017 – May 2021